Practice workbook · Sum and product of roots

Writing down alpha + beta and alpha*beta from the coefficients and using them to find expressions such as alpha^2 + beta^2, alpha^3 + beta^3, alpha^4 + beta^4 or fractions in the roots, without solving the equation.

21 question parts: 1 worked-example retry and 20 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-07

1. Question 11 · part b.i · 2 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part b.i only.

11 The roots of a quadratic equation E are α and β where $\alpha > \beta > 0$

Given that $\alpha - \beta = 2\sqrt{6}$ and $\alpha^2 + \beta^2 = 30$

(a) show that

(i) $\alpha\beta = 3$

(4)

(ii) $\alpha + \beta = 6$

(2)

(b) Without solving E

(i) find the value of $\alpha^4 + \beta^4$

(2)

(ii) find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\alpha^4 = P + Q\sqrt{6}$ where P and Q are positive integers,

(c) find the value of P and the value of Q

(2)

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1. Question 11 · part b.i · 2 marks · Working and answer

Working and answer

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2. Question 10 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

10 The roots of a quadratic equation are α and β where

$$\alpha+\beta=-\frac{5}{2} \text{ and } \alpha^3+\beta^3=\frac{115}{8}$$

(a) Show that $\alpha\beta=4$

(3)

(b) Form a quadratic equation with integer coefficients, that has roots

$$\frac{\alpha^2+1}{\beta} \text{ and } \frac{\beta^2+1}{\alpha}$$

(7)

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2. Question 10 · part a · 3 marks · Working and answer

Working and answer

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3. Question 5 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

5 (a) Show that $(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)=\alpha^3+\beta^3$

(2)

The quadratic equation $2x^2-6x-7=0$ has roots α and β

Without solving the equation

(b) form a quadratic equation, with integer coefficients, which has roots $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$

(6)

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3. Question 5 · part a · 2 marks · Working and answer

Working and answer

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4. Question 4 · part a · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 The quadratic equation

$$2x^2 + 4x + 3 = 0$$

has roots α and β

(a) Without solving the equation, show that $\alpha^2 + \beta^2 = 1$

(4)

(b) Without solving the equation, find the value of $\alpha^4 + \beta^4$

(3)

(c) Hence form a quadratic equation with integer coefficients that has roots α^4 and β^4

(3)

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4. Question 4 · part a · 4 marks · Working and answer

Working and answer



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5. Question 4 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

4 The quadratic equation

$$2x^2 + 4x + 3 = 0$$

has roots α and β

(a) Without solving the equation, show that $\alpha^2 + \beta^2 = 1$

(4)

(b) Without solving the equation, find the value of $\alpha^4 + \beta^4$

(3)

(c) Hence form a quadratic equation with integer coefficients that has roots α^4 and β^4

(3)

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5. Question 4 · part b · 3 marks · Working and answer

Working and answer

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6. Question 8 · part a · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 The quadratic equation $3x^2 - kx - 1 = 0$, where k is a positive integer, has roots α and β

(a) Show that $\alpha^2 + \beta^2 = \dfrac{k^2 + 6}{9}$

(3)

Given that $\alpha^4 + \beta^4 = \dfrac{466}{81}$

(b) find the value of k

(5)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha^3 + \beta}{\beta} \text{ and } \frac{\beta^3 + \alpha}{\alpha}$$

(6)

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6. Question 8 · part a · 3 marks · Working and answer

Working and answer

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7. Question 10 · part a.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part a.i only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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7. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 10 · part a.ii · 5 marks

2019 JUN · Paper 02

Further practice · Attempt part a.ii only.

10 The roots of the equation $x^2 + 3x - 5 = 0$ are α and β.

(a) Without solving the equation, find

(i) the value of $\alpha^2 + \beta^2$

(ii) the value of $\alpha^4 + \beta^4$

(5)

Given that $\alpha > \beta$ and without solving the equation

(b) show that $\alpha - \beta = \sqrt{29}$

(2)

(c) Factorise $\alpha^4 - \beta^4$ completely.

(3)

(d) Hence find the exact value of $\alpha^4 - \beta^4$

(2)

Given that $\beta^4 = p + q\sqrt{29}$ where p and q are positive constants

(e) find the value of p and the value of q.

(3)

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8. Question 10 · part a.ii · 5 marks · Working and answer

Working and answer

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9. Question 8 · part b · 4 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

8 (a) Using the binomial expansion, or otherwise, find the complete expansion of

$$(x + y)^3$$

(1)

The quadratic equation

$$2x^2 + 3x + 4 = 0$$

has roots α and β

(b) Without solving the equation, find the value of

$$\alpha^3 + \beta^3$$

(4)

(c) Hence, form a quadratic equation with integer coefficients that has roots

$\frac{\alpha}{\beta^2}$ and $\frac{\beta}{\alpha^2}$

(5)

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9. Question 8 · part b · 4 marks · Working and answer

Working and answer

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10. Question 7 · part a · 1 mark

2025 NOV · Paper 01

Further practice · Attempt part a only.

7 (a) Show that $(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3+\beta^3$

(1)

The equation $2x^2+8x-k=0$ has roots α and β and where k is a constant

Given that $\alpha^3+\beta^3=-94$

(b) show that $k=5$

(4)

Given that $\alpha>\beta$ and without solving the equation $2x^2+8x-k=0$

(c) (i) show that $\alpha-\beta=\sqrt{26}$

(3)

(ii) hence find the exact value of $\alpha^3-\beta^3$

(2)

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10. Question 7 · part a · 1 mark · Working and answer

Working and answer

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11. Question 7 · part c.i · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part c.i only.

7 (a) Show that $(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3+\beta^3$

(1)

The equation $2x^2+8x-k=0$ has roots α and β and where k is a constant

Given that $\alpha^3+\beta^3=-94$

(b) show that $k=5$

(4)

Given that $\alpha>\beta$ and without solving the equation $2x^2+8x-k=0$

(c) (i) show that $\alpha-\beta=\sqrt{26}$

(3)

(ii) hence find the exact value of $\alpha^3-\beta^3$

(2)

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11. Question 7 · part c.i · 3 marks · Working and answer

Working and answer

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12. Question 7 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.

7 (a) Show that $(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3+\beta^3$

(1)

The equation $2x^2+8x-k=0$ has roots α and β and where k is a constant

Given that $\alpha^3+\beta^3=-94$

(b) show that $k=5$

(4)

Given that $\alpha>\beta$ and without solving the equation $2x^2+8x-k=0$

(c) (i) show that $\alpha-\beta=\sqrt{26}$

(3)

(ii) hence find the exact value of $\alpha^3-\beta^3$

(2)

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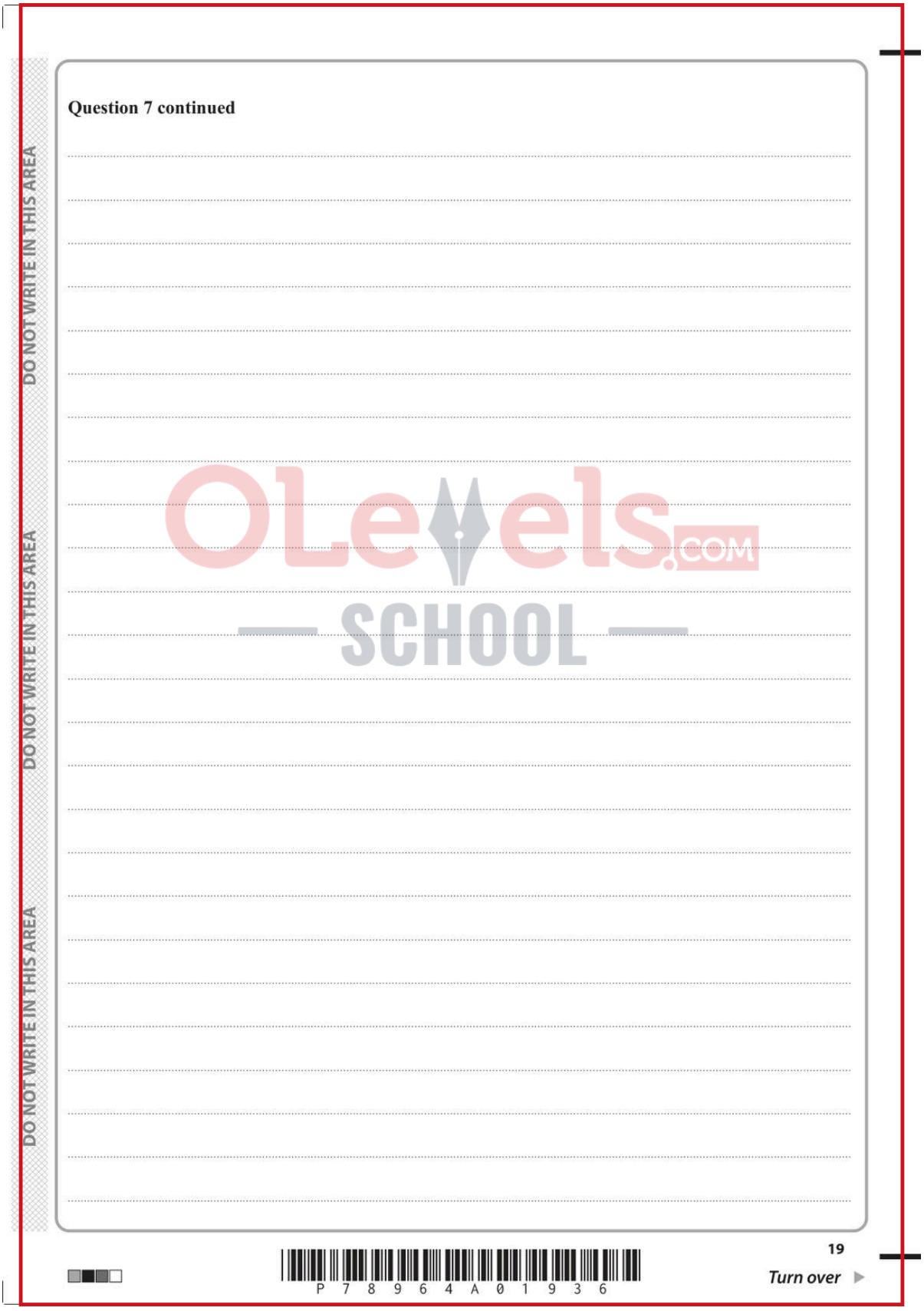
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12. Question 7 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.



Question 7 continued

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12. Question 7 · part c.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part c.ii only.

Question 7 continued

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12. Question 7 · part c.ii · 2 marks

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Further practice · Attempt part c.ii only.

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Question 7 continued

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(Total for Question 7 is 10 marks)

12. Question 7 · part c.ii · 2 marks · Working and answer

Working and answer

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13. Question 5 · part a · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

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13. Question 5 · part a · 2 marks · Working and answer

Working and answer

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14. Question 5 · part b · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

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14. Question 5 · part b · 2 marks · Working and answer

Working and answer

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15. Question 5 · part c · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

5 (a) Show that $(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \alpha^3 + \beta^3$

(2)

The roots of the equation $2x^2 + 3x + 6 = 0$ are α and β

Without solving the equation,

(b) find the value of $\alpha^3 + \beta^3$

(2)

(c) Show that $(\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = \alpha^4 + \beta^4$

(2)

(d) Form a quadratic equation with integer coefficients that has roots $(\alpha^3 - \beta)$ and $(\beta^3 - \alpha)$

(6)

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15. Question 5 · part c · 2 marks · Working and answer

Working and answer

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16. Question 10 · part d · 1 mark

2024 NOV · Paper 01

Further practice · Attempt part d only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

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16. Question 10 · part d · 1 mark · Working and answer

Working and answer

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17. Question 5 · part a.i · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a.i only.

5 The roots of the quadratic equation $2x^2 + (6 + 2p)x + 2p = 0$ are α and β

(a) Write down an expression in terms of p for

(i) $\alpha + \beta$ (ii) $\alpha\beta$

(2)

(b) Show that $(\alpha - \beta)^2 = 9 + 2p + p^2$

(4)

Given that $(\alpha - \beta) = 3$

(c) find the possible values of p

(3)

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17. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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18. Question 8 · part b.i · 0 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

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18. Question 8 · part b.i · 0 marks · Working and answer

Working and answer

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19. Question 8 · part b.ii · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

8 $f(x) = 2x^2 - 4x + 9$

The curve C has equation $y = f(x)$
The line l with equation $y - 8x + 9 = 0$ is a tangent to C
The line k is perpendicular to l and is also the tangent to C at the point where $x = p$

(a) Find the value of p

(5)

(b) Show that

(i) $(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$

(ii) $\alpha^4 + \beta^4 = \left((\alpha + \beta)^2 - 2\alpha\beta\right)^2 - 2(\alpha\beta)^2$

(2)

The quadratic equation $f(x) = 0$ has roots α and β
Without solving the equation and using your results from part (b)

(c) form a quadratic equation with integer coefficients, that has roots

$\alpha^3 - \beta$ and $\beta^3 - \alpha$

(8)

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19. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

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20. Question 11 · part a.i · 0 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

11 The quadratic equation $x^2 - px + q = 0$ where $p > 0$, has roots α and β.

Given that $2\alpha\beta = 3$ and that $4(\alpha^2 + \beta^2) = k^2 - 6k - 3$ where $k > 3$

(a) (i) write down the value of q,

(ii) find an expression, in terms of k, for p.

(5)

Given also that $7\alpha\beta = 3(\alpha + \beta)$

(b) find the value of k.

(2)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha}{\alpha + \beta} \text{ and } \frac{\beta}{\alpha + \beta}$$

(5)

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20. Question 11 · part a.i · 0 marks · Working and answer

Working and answer

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21. Question 8 · part a · 1 mark

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8 (a) Using the binomial expansion, or otherwise, find the complete expansion of

$$(x + y)^3$$

(1)

The quadratic equation

$$2x^2 + 3x + 4 = 0$$

has roots α and β

(b) Without solving the equation, find the value of

$$\alpha^3 + \beta^3$$

(4)

(c) Hence, form a quadratic equation with integer coefficients that has roots

$\frac{\alpha}{\beta^2}$ and $\frac{\beta}{\alpha^2}$

(5)

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21. Question 8 · part a · 1 mark · Working and answer

Working and answer

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